BIO 2122 | Anatomy & Physiology II | Fall 2026

Lecture 11

Urinary System I



The Nephron and Glomerular Filtration

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Scioto Community College, North Campus

Monday, October 5, 2026 | Science Hall S-214

Unit 2 | Last lecture before the Unit 2 Exam (Wednesday, October 7)

Notes

Tonight we begin the urinary system. This is the last lecture before the Unit 2 Exam on Wednesday, October 7, and the material in this deck is on that exam.

Urinary System II (tubular reabsorption, tubular secretion and the concentration of urine) begins with Lecture 12 on Monday, October 12.

Learning Objectives

After tonight, you should be able to:

1. Name the parts of the nephron in order and locate each part in the kidney.
2. Compare cortical and juxtamedullary nephrons.
3. Trace blood from the afferent arteriole through the glomerulus and the efferent arteriole to the peritubular capillaries or vasa recta.
4. Describe the three layers of the filtration membrane and what each layer holds back.
5. Calculate net filtration pressure (NFP) from HP_{gc}, OP_{gc} and HP_{cs}.
6. Define glomerular filtration rate (GFR), explain GFR = K_f × NFP, and state the normal GFR.
7. Explain the intrinsic and extrinsic controls of GFR.

Exam topic: Nephron anatomy
Objectives 1 to 4

Exam topic: Glomerular filtration & GFR regulation
Objectives 5 to 7

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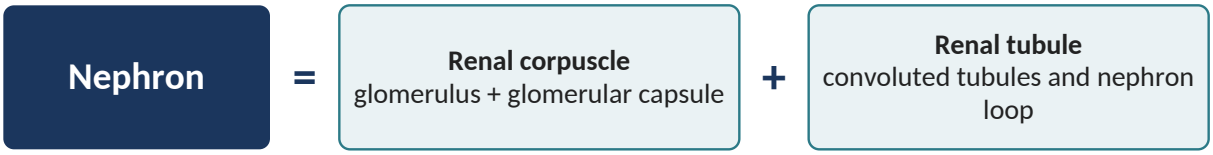
Notes

Objectives 1 through 4 are the exam topic Nephron anatomy. Objectives 5 through 7 are the exam topic Glomerular filtration & GFR regulation.

Use the terms exactly as they appear on these slides. On an exam, name a structure in full the first time, then abbreviate.

The Nephron: Functional Unit of the Kidney

- Each kidney contains **about 1 million nephrons**.
- Each nephron has two parts:
  - **Renal corpuscle:** where blood plasma is filtered (slide 5)
  - **Renal tubule:** where the filtrate is processed into urine (slide 6)
- Nephrons drain into **collecting ducts**. Each collecting duct receives filtrate from many nephrons.
- The kidneys receive about one quarter of the resting cardiac output, about 1,200 mL of blood per minute.



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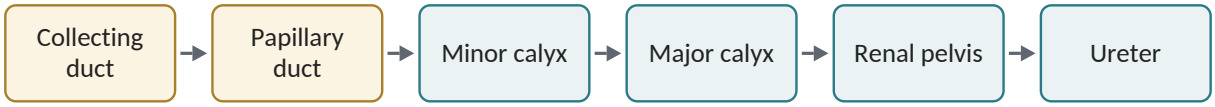
Keep the count in mind: about 1 million nephrons in each kidney. The glomerular filtration rate on slide 19 is the combined output of all of them, in both kidneys.

The kidneys receive far more blood than their size would suggest. That blood flow is what makes a high filtration rate possible.

Where the Parts Sit: Cortex and Medulla

- **Renal cortex** (outer region): every renal corpuscle, every **proximal convoluted tubule (PCT)** and every **distal convoluted tubule (DCT)**
- **Renal medulla** (renal pyramids): the **nephron loops** and the **collecting ducts**, which give the pyramids their striped look
- Cortical tissue that extends between the pyramids forms the **renal columns**.
- After the collecting ducts, the fluid is urine. It drains along the path below and is not changed again.

Urine path



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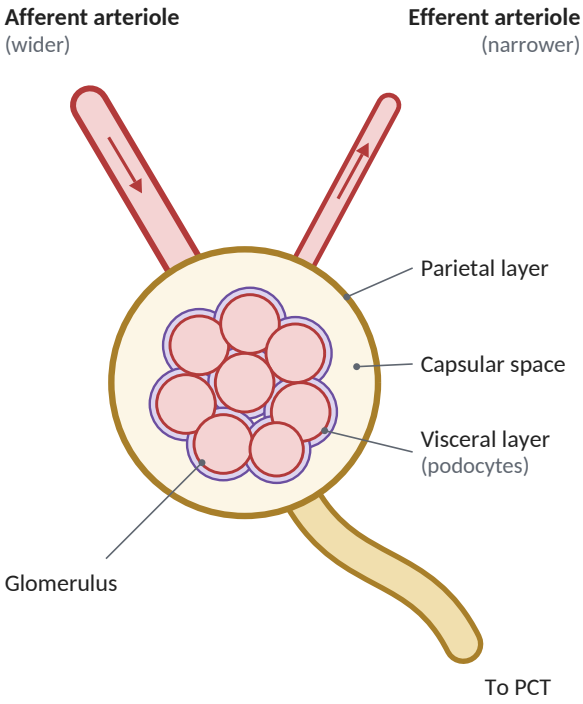
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Every renal corpuscle and every convoluted tubule sits in the cortex. The medulla holds the nephron loops and the collecting ducts.

Beyond the papillary ducts, the fluid is urine. The calyces, the renal pelvis and the ureter carry it but do not change its composition.

The Renal Corpuscle

- **Renal corpuscle** = glomerulus + glomerular capsule
- **Glomerulus:** a tuft of fenestrated capillaries, fed by the afferent arteriole and drained by the efferent arteriole
- **Glomerular capsule:** a cup-shaped, hollow structure around the glomerulus
  - **Parietal layer:** simple squamous epithelium; structural only
  - **Visceral layer:** podocytes that cling to the glomerular capillaries
  - **Capsular space:** between the two layers; receives the filtrate
- Some sources say Bowman's capsule. On exams, write **glomerular capsule**.



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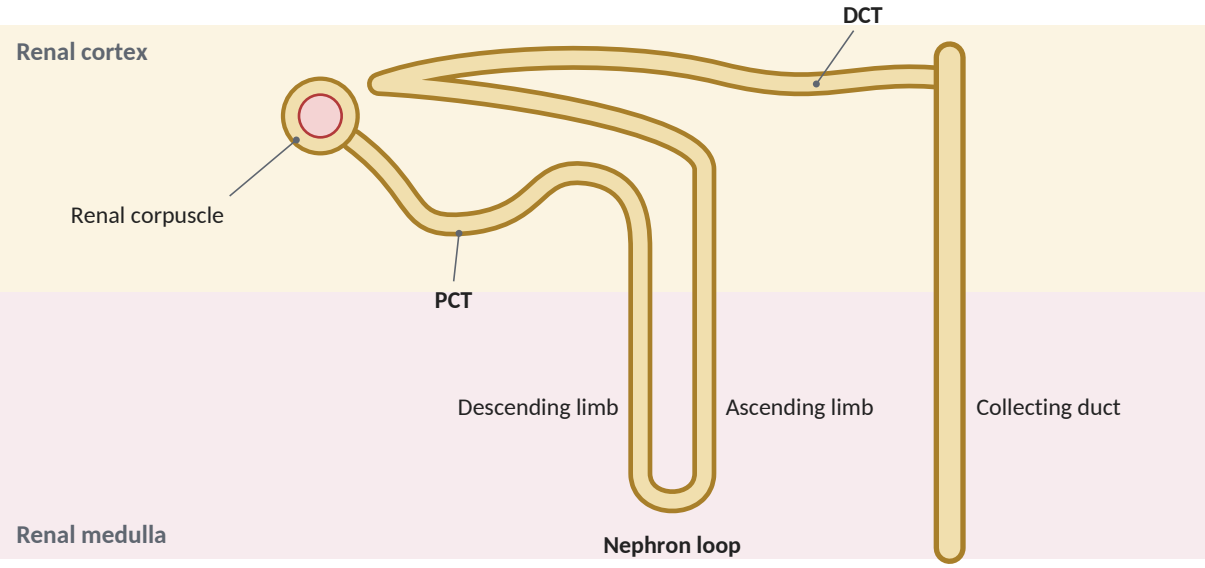
The parietal layer is structural only; it does not take part in forming filtrate. The visceral layer is made of podocytes, and their foot processes are part of the filtration membrane (slide 12).

Filtrate collects in the capsular space and flows from there into the PCT.

The Renal Tubule, in Order

Glomerular capsule → **PCT** → Nephron loop: descending limb, then ascending limb → **DCT** → Collecting duct

The collecting duct receives filtrate from many nephrons. Older name for the nephron loop: loop of Henle. In this course, write **nephron loop**.



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Notes

Learn the sequence as a sentence: filtrate leaves the glomerular capsule, passes through the PCT, goes down the descending limb and up the ascending limb of the nephron loop, passes through the DCT, and enters a collecting duct.

Many nephrons share one collecting duct, so the collecting duct is not part of a single nephron.

Renal Tubule Segments: Structure Fits Function

Segment	Epithelium	What to remember
Proximal convoluted tubule (PCT)	Cuboidal cells with dense microvilli (brush border) and many mitochondria	Reabsorbs most of the filtrate (Lecture 12)
Nephron loop: descending limb	Mostly simple squamous epithelium (thin segment)	Carries the filtrate down into the medulla
Nephron loop: ascending limb	Thick segment of cuboidal to low columnar cells in most nephrons	Its last part forms the macula densa (slide 20)
Distal convoluted tubule (DCT)	Cuboidal cells with few microvilli	Fine-tunes reabsorption under hormone control (Unit 3)
Collecting duct	Principal cells and intercalated cells	Principal cells: water and Na^+ balance. Intercalated cells: acid-base balance.

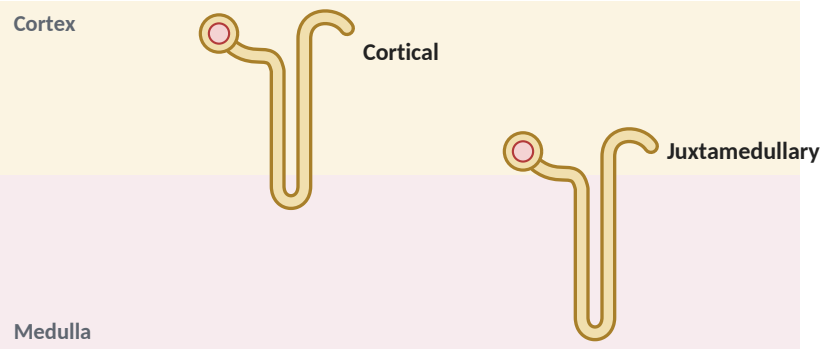
Notes

Read the cell structure as a clue to function. Dense microvilli and many mitochondria in the PCT mean heavy, energy-dependent reabsorption.

We follow reabsorption segment by segment in Lecture 12. The principal cells return in Lecture 13.

Cortical vs. Juxtamedullary Nephrons

Feature	Cortical nephrons	Juxtamedullary nephrons
Share of all nephrons	About 85%	About 15%
Renal corpuscle	Outer cortex	Deep cortex, close to the cortex-medulla junction
Nephron loop	Short; dips only into the outer medulla	Long; reaches deep into the medulla
Efferent arteriole feeds	Peritubular capillaries	Vasa recta
Concentrating urine	Small role	Key role (Lecture 13)



Corpuscle position predicts loop length: the deeper the corpuscle, the longer the nephron loop.

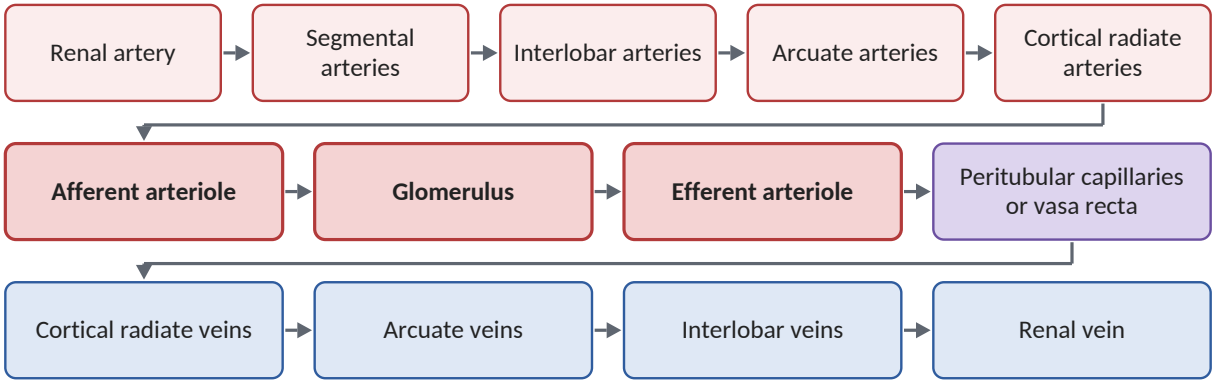
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Notes

Most nephrons are cortical. The juxtamedullary nephrons are fewer, but their long nephron loops and the vasa recta are what allow the kidney to produce concentrated urine. We return to that in Lecture 13.

Blood Supply: Into and Out of the Glomerulus

- The **afferent arteriole** brings blood into the glomerulus.
- The **efferent arteriole** carries blood out of the glomerulus. It is **narrower** than the afferent arteriole.
- The glomerulus is the only capillary bed that is both fed and drained by arterioles. The narrow efferent arteriole resists outflow, so pressure in the glomerulus stays high. That pressure drives filtration.



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Notes

Follow the arrows from the renal artery to the renal vein. The step that matters tonight is in the middle row: blood enters the glomerulus through the afferent arteriole and leaves through the narrower efferent arteriole.

That resistance on the outflow side keeps glomerular capillary pressure high, and that pressure is what drives filtration.

Peritubular Capillaries and Vasa Recta

Peritubular capillaries

- Arise from the efferent arterioles of **cortical nephrons**
- Cling to the renal tubules **in the cortex**
- Low pressure and porous, so they readily take up the water and solutes reabsorbed by the tubules

Vasa recta

- Arise from the efferent arterioles of **juxtamedullary nephrons**
- Long, straight vessels that run **alongside the long nephron loops in the medulla**
- Help the kidney form concentrated urine (Lecture 13)

Pressure along the way

Afferent arteriole → **Glomerulus:** high pressure, so fluid leaves (filtration) → **Efferent arteriole:** resists outflow → **Peritubular capillaries:** low pressure, so reabsorbed fluid moves in

Both capillary beds come after the glomerulus, so the blood in them has already been filtered.

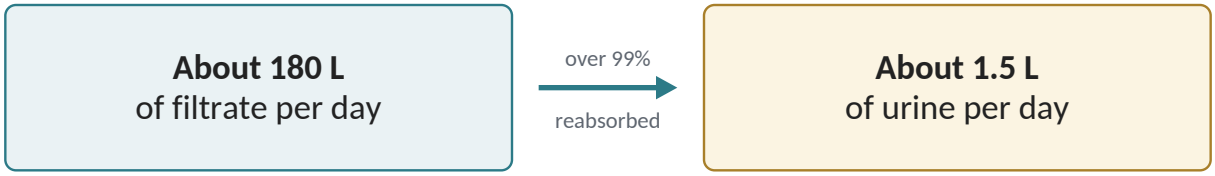
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Notes

Both capillary beds receive blood that has already passed through the glomerulus. Their low pressure is an advantage: fluid moves into them easily, which is why they can reclaim the water and solutes that the tubules reabsorb.

Urine Formation Begins with Glomerular Filtration

- Glomerular filtration is **passive**: bulk flow driven by hydrostatic pressure. The kidney spends no metabolic energy on filtration itself.
- It is **nonselective**: anything small enough to cross the filtration membrane is filtered, useful or not.
- **Filtrate** = blood plasma minus most plasma proteins.
- The tubules then reclaim what the body needs (tubular reabsorption) and add what it must remove (tubular secretion): Lecture 12.



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Notes

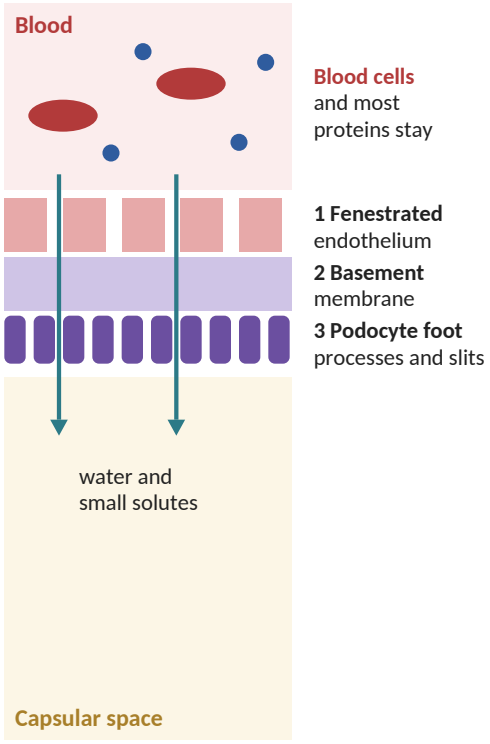
Filtration is passive and does not sort useful solutes from wastes. That is why the tubules must reclaim almost everything: of about 180 L of filtrate each day, only about 1.5 L leaves as urine.

The daily filtrate volume comes from the normal GFR on slide 19.

The Filtration Membrane

Three layers, in the order filtrate crosses them:

1. **Fenestrated endothelium** of the glomerular capillaries: pores let plasma through but hold back blood cells.
2. **Basement membrane:** fused basal laminae. Its negatively charged glycoproteins restrict most plasma proteins.
3. **Podocytes** (visceral layer of the glomerular capsule): foot processes separated by **filtration slits**, bridged by slit diaphragms. The last barrier to small proteins.



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Notes

Learn the three layers in the order filtrate crosses them: fenestrated endothelium, basement membrane, podocyte filtration slits.

The fenestrations stop blood cells. The negatively charged basement membrane restricts most proteins. The slit diaphragms are the last barrier.

What Crosses, What Stays in the Blood

Normally filtered (in the filtrate)

- Water
- Glucose and amino acids
- Ions: Na^+, K^+, Cl^-, HCO_3^-
- Nitrogenous wastes: urea, uric acid, creatinine
- Molecules smaller than about 3 nm pass freely

Normally held back (stay in the blood)

- Blood cells: erythrocytes, leukocytes, platelets
- Most plasma proteins, including albumin
- Molecules larger than about 5 nm are generally held back

Why it matters

- Proteins kept in the capillaries create the colloid osmotic pressure that opposes filtration (OP_{gc}, slide 17).
- Protein or blood cells in urine (proteinuria, hematuria) point to a damaged filtration membrane.

Notes

Plasma proteins that stay in the capillaries matter twice tonight. They keep the colloid osmotic pressure that opposes filtration (OP_{gc}, slide 17), and their presence in urine tells us the filtration membrane is damaged.

Why the Glomerulus Is an Efficient Filter

1 Large surface area

Many glomerular capillary loops in each kidney give a large filtration surface.

Part of K_f

2 High permeability

The filtration membrane is far more permeable to water and small solutes than other capillary walls.

Part of K_f

3 High pressure

Glomerular capillary pressure is higher than in other capillary beds, because the efferent arteriole is narrower than the afferent arteriole.

Part of NFP

All three features appear in **GFR = K_f × NFP**. Surface area and permeability make up the filtration coefficient, K_f (slide 18). Pressure is the net filtration pressure, NFP (slide 17).

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Notes

Each feature maps onto a term in the GFR equation. Surface area and permeability are the filtration coefficient, K_f (slide 18). Pressure is the net filtration pressure, NFP (slide 17).

Two Kinds of Pressure at the Filtration Membrane

- **Hydrostatic pressure (HP):** the pressure of a fluid pushing against a wall. It pushes fluid out of the compartment it is in.
- **Colloid osmotic pressure (OP):** the osmotic pull of proteins that cannot cross the membrane. It pulls water toward the side that has the proteins.

Promotes filtration (outward)	Hydrostatic pressure in the glomerular capillaries (HP_{gc})
Opposes filtration (inward)	Colloid osmotic pressure in the glomerular capillaries (OP_{gc}) Hydrostatic pressure in the capsular space (HP_{cs})

Subscripts

gc = glomerular capillary

cs = capsular space

Colloid osmotic pressure in the capsular space is essentially zero, because the filtrate contains almost no protein. That is why the NFP equation has no OP term for the capsular space.

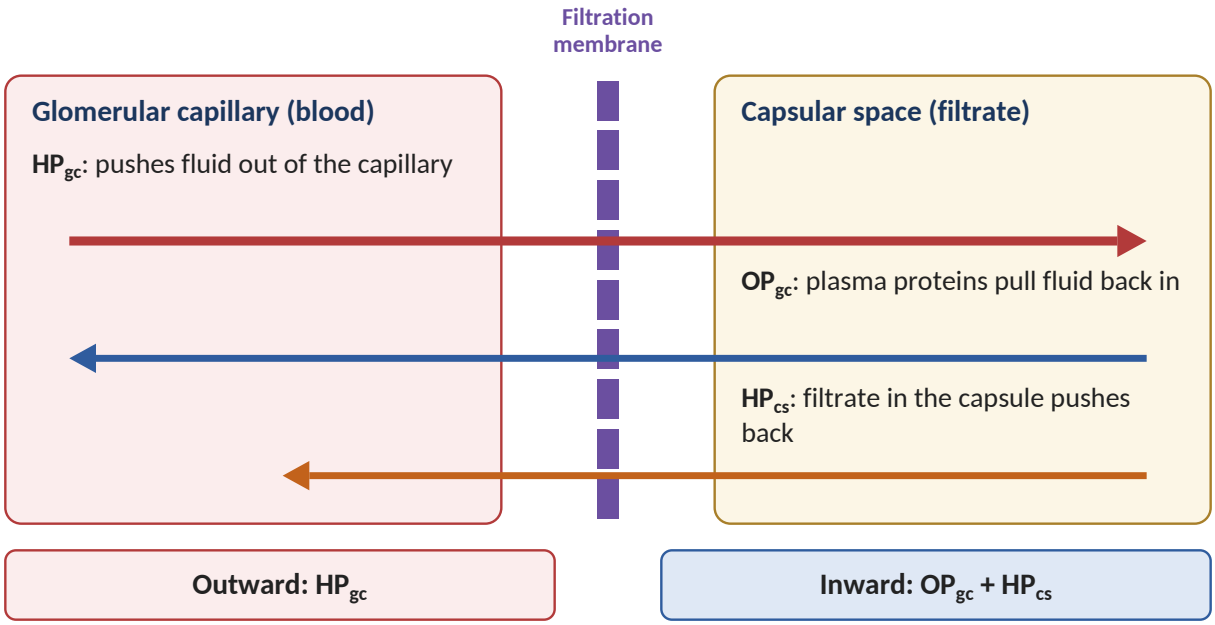
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Notes

The subscripts tell you where a pressure is measured: gc means glomerular capillary and cs means capsular space.

Hydrostatic pressure pushes fluid out of the compartment it is in. Colloid osmotic pressure pulls water toward the side with the proteins.

The Forces Diagram



NFP = the outward pressure minus the sum of the inward pressures (slide 17)

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Notes

This is the diagram I draw on the board before every NFP problem. One arrow points out of the capillary: HP_{gc}. Two arrows point back toward the capillary: OP_{gc} and HP_{cs}.

Every arrow that points back toward the capillary goes inside the parentheses on slide 17.

Net Filtration Pressure (NFP)

$$NFP = HP_{gc} - (OP_{gc} + HP_{cs})$$

all pressures in mm Hg

What each term means

- **HP_{gc}**, glomerular capillary hydrostatic pressure: pushes fluid out of the capillary. *Promotes filtration.*
- **OP_{gc}**, glomerular capillary colloid osmotic pressure: plasma proteins pull fluid back into the capillary. *Opposes filtration.*
- **HP_{cs}**, capsular space hydrostatic pressure: filtrate already in the capsule pushes back. *Opposes filtration.*

Set it up the way we do on the board

1. Write the equation with the parentheses.
2. Add the two opposing pressures inside the parentheses.
3. Subtract that sum from HP_{gc}.
4. Report NFP in mm Hg.

Board example: HP_{gc} = 60 mm Hg, OP_{gc} = 32 mm Hg, HP_{cs} = 16 mm Hg
NFP = 60 - (32 + 16) = 60 - 48 = **12 mm Hg**

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Notes

Set up every NFP problem the same way. Write the equation with the parentheses first. Add the two opposing pressures inside the parentheses, then subtract that sum from HP_{gc}. Both opposing pressures go inside the parentheses; neither one is added to HP_{gc}.

Check your setup against the forces diagram on slide 16, and always report NFP in mm Hg. The board example uses numbers chosen for practice.

Filtration Coefficient (K_f)

K_f depends on two properties of the filtration membrane:

1. **Total surface area** available for filtration
2. **Permeability** of the filtration membrane

- Larger surface area or a more permeable membrane → larger K_f → more filtrate at the same NFP.
- Mesangial cells inside the glomerulus can contract and reduce the surface area available for filtration.
- Diseases that scar or thicken the filtration membrane lower K_f, so GFR falls even when NFP is normal.
- K_f is not a pressure. Its units are mL/min of filtrate for each mm Hg of NFP.

Same NFP, different K_f

Healthy filtration membrane	**Scarred glomeruli**
normal K_f → normal GFR	lower K_f → lower GFR

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Notes

Think of K_f as how much filtrate the membrane passes for each mm Hg of net filtration pressure.

In healthy kidneys K_f changes little from minute to minute. Disease that scars the glomeruli is what usually lowers it.

Glomerular Filtration Rate (GFR)

Glomerular filtration rate (GFR): the volume of filtrate formed each minute by all the nephrons in both kidneys

$$GFR = K_f \times NFP$$

- Normal GFR: **120 to 125 mL/min**
- GFR is directly proportional to NFP: what raises NFP raises GFR, and what lowers NFP lowers GFR.
- Because NFP is small, a small change in HP_{gc} causes a large change in GFR.
- Three factors govern GFR: NFP, total surface area and membrane permeability. The last two make up K_f.

Per day

125 mL/min × 1,440 min/day

= 180,000 mL/day

≈ 180 L of filtrate

every day

Notes

GFR is directly proportional to NFP. Any change that raises NFP raises GFR, and any change that lowers NFP lowers GFR.

Because NFP is small, a modest change in glomerular capillary pressure produces a large change in GFR. That is why the body regulates the afferent arteriole so closely (slides 21 to 24).

The Juxtaglomerular Complex (JG Complex)

The **juxtaglomerular complex (JG complex)** is where the last part of the ascending limb of the nephron loop lies against the afferent arteriole of the same nephron.

Cell type	Location	Role
Macula densa	Tall, closely packed cells in the wall of the ascending limb	Chemoreceptors: sense NaCl in the filtrate
Granular cells (juxtaglomerular cells)	Enlarged smooth muscle cells in the wall of the afferent arteriole	Mechanoreceptors: sense blood pressure by stretch. Release renin.
Extraglomerular mesangial cells	Between the arteriole and the tubule	May pass signals between the macula densa and the granular cells

Two sensors, two pathways

Macula densa senses NaCl → tubuloglomerular feedback (slide 22)

Granular cells sense blood pressure → renin release (slide 24)

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Notes

The JG complex is where the tubule reports back to its own blood supply. The macula densa senses the filtrate, and the granular cells sense blood pressure and release renin.

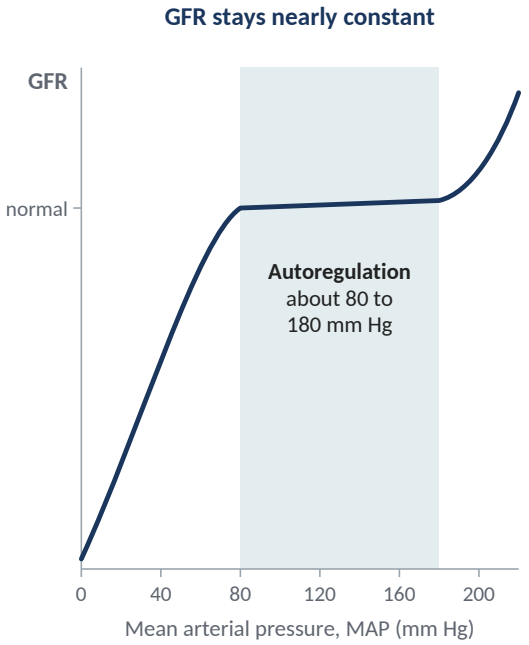
Keep the two sensors separate in your notes; we use both of them on the next four slides.

Intrinsic Controls: Renal Autoregulation

- **Goal:** keep GFR nearly constant while systemic blood pressure (BP) changes
- Works while mean arterial pressure (MAP) is **about 80 to 180 mm Hg**

Two intrinsic mechanisms

1. **Myogenic mechanism:** vascular smooth muscle contracts when it is stretched
   - BP rises → afferent arteriole is stretched → it **constricts** → HP_{gc} and GFR stay near normal
   - BP falls → afferent arteriole **dilates** → HP_{gc} and GFR stay near normal
2. **Tubuloglomerular feedback:** the macula densa senses NaCl in the filtrate and adjusts the afferent arteriole (slide 22)



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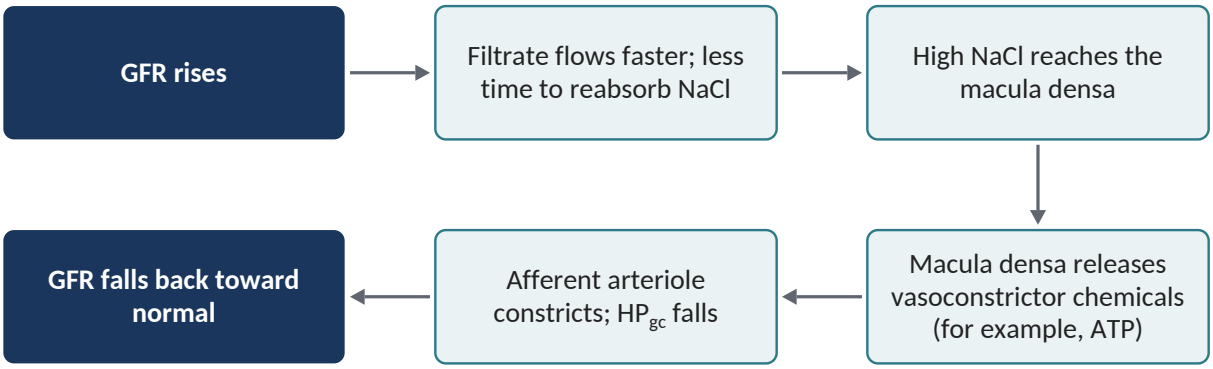
Notes

Both intrinsic mechanisms act on the afferent arteriole. Constricting the afferent arteriole lets less blood into the glomerulus, so HP_{gc} falls; dilating it does the opposite.

Autoregulation keeps GFR nearly constant only while MAP stays between about 80 and 180 mm Hg. Below about 80 mm Hg, autoregulation can no longer protect GFR.

Tubuloglomerular Feedback

Sensor: macula densa cells in the ascending limb of the nephron loop sense the NaCl concentration of the filtrate.



Low NaCl at the macula densa has the opposite effect: the afferent arteriole dilates and GFR rises. The response removes the stimulus, so this is **negative feedback**.

Compare: the myogenic mechanism responds to stretch of the afferent arteriole; tubuloglomerular feedback responds to NaCl at the macula densa. Both act on the afferent arteriole.

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Notes

Tubuloglomerular feedback is a negative feedback loop: the response, a lower GFR, removes the stimulus, high NaCl at the macula densa.

Draw the loop on paper until you can explain each arrow in one sentence.

Extrinsic Controls of GFR

Extrinsic controls: the sympathetic nervous system and the renin-angiotensin-aldosterone mechanism. Their job is to protect systemic blood pressure, even at the cost of GFR.

Sympathetic nervous system

- At rest, renal blood vessels are dilated and autoregulation is in charge.
- In extreme stress or blood loss, norepinephrine from sympathetic nerves and epinephrine from the adrenal medulla **constrict the afferent arterioles**.
- GFR falls, blood volume is conserved, and more blood reaches vital organs.
- Also stimulates renin release (slide 24).

Renin-angiotensin-aldosterone mechanism

- The main hormonal way the body raises blood pressure
- Starts when the granular cells release renin (slide 24)
- Angiotensin II constricts arterioles throughout the body, which raises systemic BP.

Strong sympathetic stimulation overrides autoregulation: the body accepts a lower GFR to keep systemic blood pressure up.

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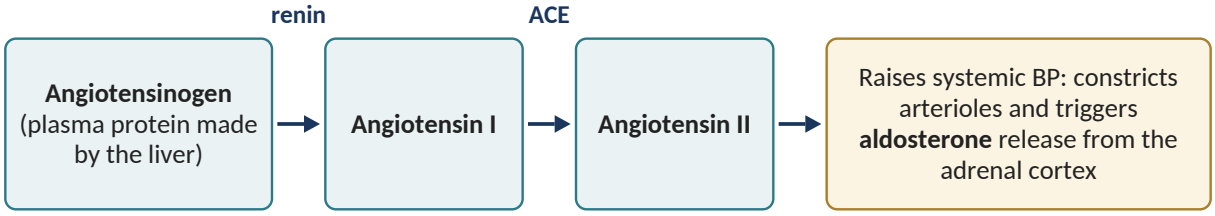
The extrinsic controls put the whole body ahead of the kidney. When blood pressure falls sharply, maintaining systemic pressure matters more than maintaining GFR, so the afferent arterioles constrict even though filtration drops.

Renin Release from Granular Cells

Renin is released by the **granular cells** of the JG complex. Three triggers:

1. **Sympathetic stimulation** (β_1-adrenergic receptors on the granular cells)
2. **Low NaCl at the macula densa**
3. **Less stretch of the granular cells** (low blood pressure in the afferent arteriole)

ACE = angiotensin converting enzyme, found mainly in the capillary endothelium of the lungs



Net effect: a higher systemic BP restores blood flow to the kidneys and, with it, GFR.

Notes

Know who releases renin, the granular cells of the JG complex, and the three triggers for its release.

The full renin-angiotensin-aldosterone pathway, including how aldosterone changes sodium reabsorption, returns in Unit 3 with fluid and electrolyte balance.

Clinical Connections: Reasoning with NFP

Situation	What changes first	Effect
Severe blood loss	Sympathetic nerves constrict the afferent arterioles: HP_{gc} ↓	NFP ↓, GFR ↓ (protects blood volume)
Kidney stone blocking a ureter	Urine backs up: HP_{cs} ↑	NFP ↓, GFR ↓
Very low plasma protein (liver disease, starvation)	OP_{gc} ↓	NFP ↑, GFR ↑
Damaged filtration membrane (glomerulonephritis)	Proteins and blood cells cross the membrane	Proteinuria and hematuria; GFR falls as glomeruli scar

Chronic kidney disease: GFR below 60 mL/min/1.73 m^2 for 3 months or more.

How to reason through a GFR question

1. Which pressure changes first?
2. Does it promote filtration (HP_{gc}) or oppose it (OP_{gc}, HP_{cs})?
3. Follow the change through NFP to GFR.

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Notes

For each case, name the pressure that changes first, then follow it through NFP to GFR. This is the reasoning I expect on exam questions that ask you to predict a change in GFR.

Summary

- **Nephron** = renal corpuscle (glomerulus + glomerular capsule) + renal tubule (PCT → nephron loop → DCT), draining into a collecting duct.
- About 85% of nephrons are cortical. About 15% are juxtamedullary, with long nephron loops.
- Blood enters the glomerulus through the afferent arteriole and leaves through the narrower efferent arteriole. Peritubular capillaries serve the cortex; vasa recta serve the medulla.
- Filtration membrane: fenestrated endothelium, basement membrane, podocyte filtration slits. It holds back blood cells and most plasma proteins.
- **NFP = HP_{gc} - (OP_{gc} + HP_{cs})**, in mm Hg. **GFR = K_f × NFP**. Normal GFR is 120 to 125 mL/min.
- Intrinsic controls: myogenic mechanism, tubuloglomerular feedback (MAP about 80 to 180 mm Hg). Extrinsic controls: sympathetic nervous system, renin-angiotensin-aldosterone mechanism.

Notes

If you can explain every line on this slide in your own words, you are ready for the nephron and filtration questions on Wednesday.

Self-Check Before Wednesday (Not Collected)

1. Starting at the glomerular capsule, name in order every part that filtrate passes through to reach the collecting duct. (slide 6)
2. Which layer of the filtration membrane holds back blood cells? Which layer is the last barrier to proteins? (slide 12)
3. Which pressure promotes filtration, and which two pressures oppose it? (slides 16 and 17)
4. Why does the NFP equation have no colloid osmotic pressure term for the capsular space? (slide 15)
5. What does the macula densa sense, and what does it signal the afferent arteriole to do when that signal is high? (slide 22)
6. Which cells release renin? Name the three triggers for renin release. (slide 24)

Notes

These questions are for self-study. They are not collected or graded. Work them without your notes first, then check yourself against the slides listed in parentheses.

Unit 2 Exam and Next Lecture

Unit 2 Exam: Wednesday, October 7, 6:00 pm, Science Hall S-214

- Covers Lectures 7 to 11 (the heart, blood vessels, the nephron and glomerular filtration), plus the carry-forward section on Unit 1 (blood and endocrine)
- Bring a pencil and your student ID.

Office hours	Monday and Wednesday, 4:30 pm to 5:45 pm, Science Hall S-305
Next lecture	Lecture 12, Monday, October 12: tubular reabsorption and secretion (Urinary System II)
Reading	Textbook Chapter 25, sections on tubular reabsorption and secretion
Questions	rsilva@sciotocc.edu

Notes

Bring a pencil and your student ID on Wednesday. If you have questions about tonight's material, come to office hours on Wednesday before the exam or email me at rsilva@sciotocc.edu.